Supplementary Information for
"Bayesian inference of the climbing grade scale"

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[Removed for blind review]

S1 Stan model

The Stan model used for all 20 analyses is shown in Listing 1. The model corresponds to the dynamic Bradley-Terry formulation in Section 3 of the main text: each ascent's success indicator is a Bernoulli with logit link in $m \cdot (C_j(t) - x_i)$, where x_i is the (observed) route grade, $C_j(t)$ is the (latent) grade of the climber at the time of the ascent, and m is the slope parameter common to all climbers in the analysis. The prior on m is a weakly informative log-normal placing its central 95% mass on the interval $m \in [0.19, 1.97]$. The prior on each climber-month grade is a normal centred on the previous month's grade with standard deviation $w = 0.5$ (the discretised Wiener process), reverting to a wide normal centred on a per-dataset grade prior μ_0 for months outside the climber's data window.

Listing 1: Stan model.

```
data {
  int<lower=1> C;              // number of climbers
  int<lower=1> N;              // number of ascents
  int<lower=1> P;              // number of pages (months)
  int<lower=1> minPage[C];     // min page for each climber
  int<lower=1> maxPage[C];     // max page for each climber
  int<lower=0,upper=1> y[N]; // ascent success/failure
  int<lower=1> page[N];        // the time block (page) of each ascent
  int<lower=1> c[N];           // the climber of each ascent
  vector[N] x;                 // route grade of each ascent
  int meanGradePrior;          // the mean of the grade prior
}
parameters {
  real climberGrade[C, P];     // latent climber grade per month
  real<lower=0.0> m;           // slope of increase in difficulty per grade increment
}
model {
  m ~ lognormal(-0.5, 0.6);    // prior on slope
  for (j in 1:C) {
    for (i in 1:minPage[j]) {
      climberGrade[j, i] ~ normal(meanGradePrior, 5);
    }
    for (i in (minPage[j]+1):maxPage[j]) {
      climberGrade[j, i] ~ normal(climberGrade[j, i-1], 0.5);
    }
    for (i in (maxPage[j]+1):P) {
      climberGrade[j, i] ~ normal(meanGradePrior, 5);
    }
  }
  for (i in 1:N) {
    y[i] ~ bernoulli_logit(m * (climberGrade[c[i], page[i]] - x[i]));
  }
}
```

All analyses were run with 4 parallel chains, 4,000 iterations per chain, of which the first 2,000 were discarded as warmup, leaving 8,000 total post-warmup draws. We monitored convergence with rank-normalised split-$\hat{R}$ and bulk/tail effective sample sizes following Vehtari et al. (2021).

S2 Cross-dataset summary

Table S1 reports, for each of the 20 analyses, the number of climbers retained (n), the number of logged ascents (N), the posterior median and 95% central credible interval for $d = e^m$, and the inclusion threshold.

The selection pipeline retains climbers passing the per-attempt minimum thresholds (set to 30 attempts and 1 explicit failure for all 20 analyses). If more than 100 climbers pass these thresholds, only the top-100 by total ascent count are kept; otherwise all eligible climbers are retained. In Table S1, n is shown in bold (**100**) for the analyses where this top-100 limit was reached, and in normal weight where all eligible climbers were used.

The distinction matters for the threshold sensitivity reported in Section 5.1 of the main text: where the limit was reached, raising the attempt threshold to ≥ 50 or ≥ 100 has no effect because the top-100 most-active climbers all log many hundreds of attempts and are unaffected by any of these thresholds; where the limit was not reached, the threshold is the active selection rule. Of the 20 analyses, 10 reached the limit (all 8 Australian analyses and both German UIAA-graded analyses) and 10 did not (all 8 New Zealand analyses and both German French-graded analyses). The slope estimates are mutually consistent across this stratification: where the limit was not reached, the slope estimates are broadly similar to those where the limit was reached at the same country/style, suggesting that the limit acts as a computational ceiling rather than a substantive analytical choice.

S3 Per-climber log-odds-ratio regression

The dynamic Bradley-Terry model in the main text assumes that the log-odds-ratio of failure is a linear function of the difference between the climber's grade and the route's grade, with slope m common to all climbers. As a simple check on this assumption, and as the source of the per-climber log-linear slope estimates quoted in Section 5.1 of the main text (0.65 for the Australia-Sport per-session data set; 0.52 for the New Zealand-Sport per-session data set), we fit an ordinary log-linear regression of log(#failures/#successes) against route grade for each climber individually, treating each climber's grade as constant across the analysis window.

Figure S1 shows the resulting per-climber regression lines for the Australia-Sport per-session data set, and Figure S2 shows the same for New Zealand-Sport per-session. The slopes are broadly consistent across climbers despite the intercepts varying by approximately 10 grades, supporting the choice of a shared slope parameter. The mean per-climber slopes (0.65 for AUS, 0.52 for NZ) are below the joint Bayesian posterior estimates (0.85 for AUS, 0.80 for NZ); we do not interpret this gap as a contradiction because the two estimators differ in important ways: the per-climber regression discards within-climber temporal structure (treating C_j as constant), pools across the entire 5-year window irrespective of activity, and uses ordinary least squares on data that is not exchangeable; the joint Bayesian model, by contrast, accounts for time-varying ability via the dynamic Bradley-Terry structure and pools information across climbers. The figures should therefore be read as a qualitative model-fit check rather than as a competing estimator.

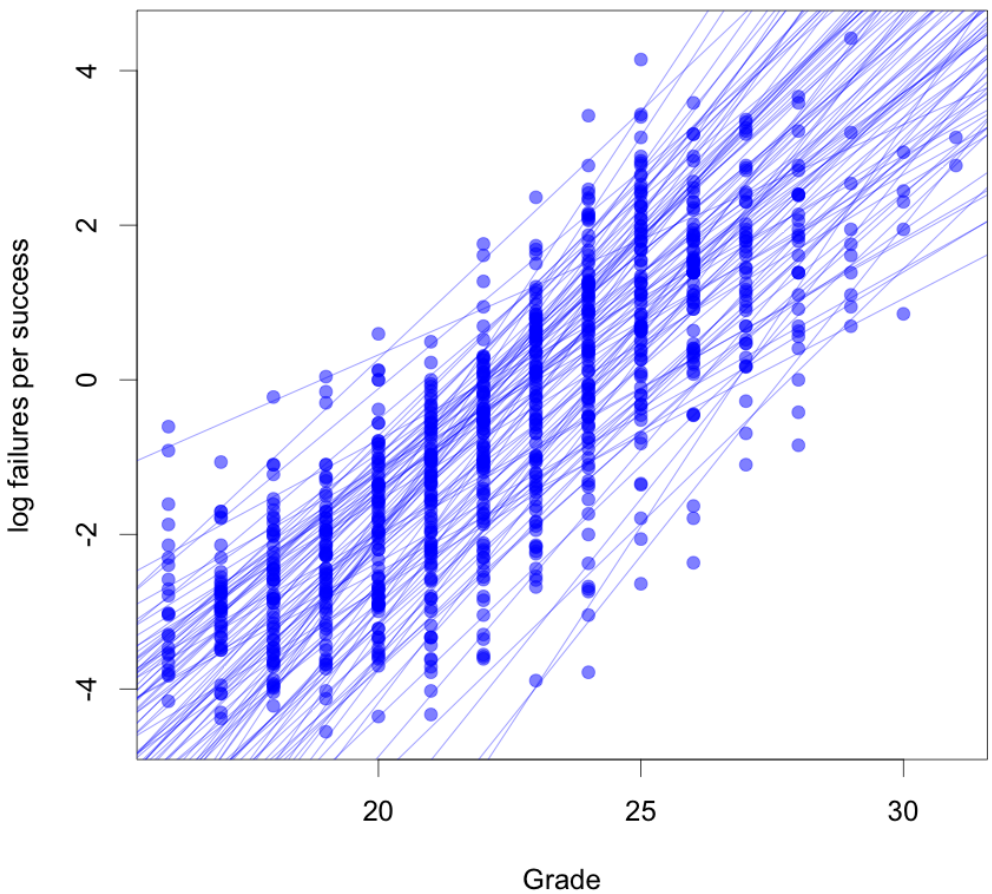

Figure S1: Per-climber log-odds-ratio of failure versus route grade for the Australia-Sport per-session data set. Each point plots log(#failures/#successes) at one grade for one climber; one regression line is fitted per climber. The slopes are visibly consistent across climbers, while the intercepts span a range of approximately 10 grades.

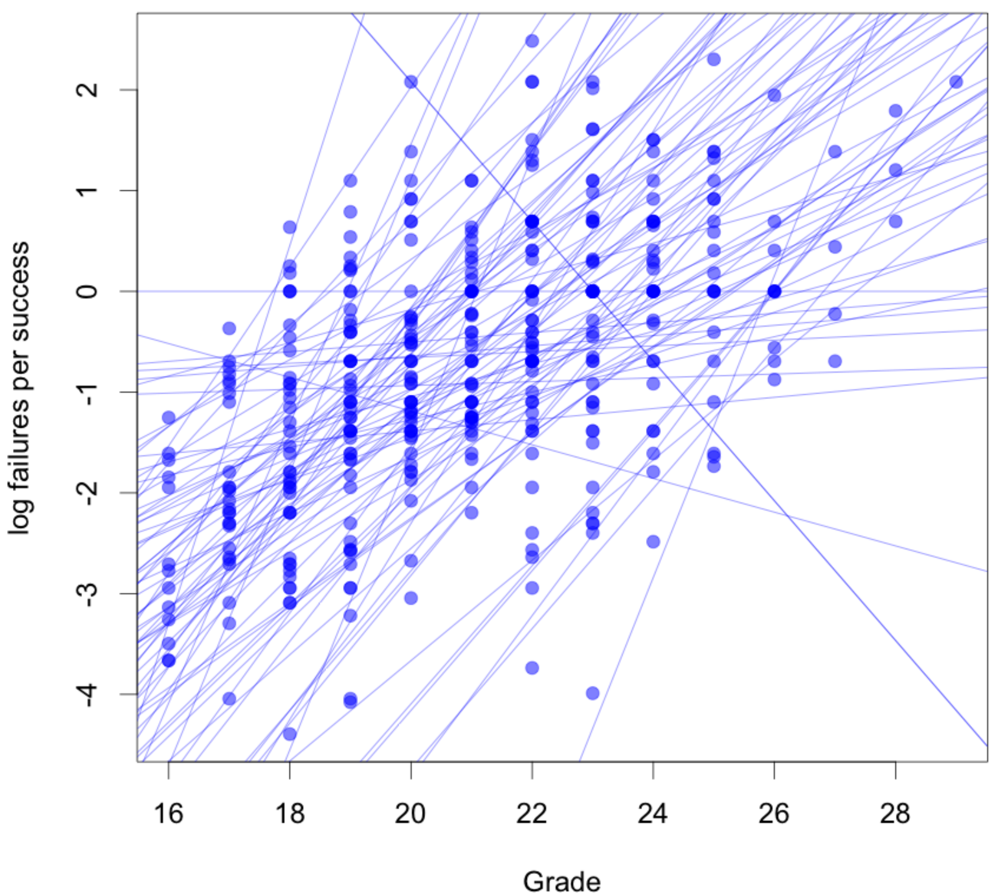

Figure S2: Per-climber log-odds-ratio of failure versus route grade for the New Zealand-Sport per-session data set. Conventions as in Figure S1.

Country	**Style**	**Grade scale**	**Game**	n	N	$d = e^m$	2.5%	97.5%	**Min. attempts**
Australia	Boulder	V-grade	attempt	**100**	25,805	2.96	2.85	3.07	≥ 30
Australia	Boulder	V-grade	session	**100**	25,344	3.03	2.91	3.15	≥ 30
Australia	Sport	Ewbank	attempt	**100**	52,947	2.24	2.21	2.28	≥ 30
Australia	Sport	Ewbank	session	**100**	48,679	2.33	2.29	2.37	≥ 30
Australia	Sport+Trad	Ewbank	attempt	**100**	61,931	2.13	2.10	2.16	≥ 30
Australia	Sport+Trad	Ewbank	session	**100**	57,512	2.18	2.15	2.21	≥ 30
Australia	Trad	Ewbank	attempt	**100**	14,929	2.02	1.96	2.09	≥ 30
Australia	Trad	Ewbank	session	**100**	14,479	2.01	1.95	2.08	≥ 30
New Zealand	Boulder	V-grade	attempt	15	1,049	3.35	2.71	4.27	≥ 30
New Zealand	Boulder	V-grade	session	15	1,027	3.35	2.68	4.32	≥ 30
New Zealand	Sport	Ewbank	attempt	89	10,027	2.19	2.11	2.27	≥ 30
New Zealand	Sport	Ewbank	session	89	9,679	2.22	2.13	2.31	≥ 30
New Zealand	Sport+Trad	Ewbank	attempt	98	11,389	2.15	2.08	2.22	≥ 30
New Zealand	Sport+Trad	Ewbank	session	98	11,024	2.17	2.09	2.26	≥ 30
New Zealand	Trad	Ewbank	attempt	8	524	1.80	1.57	2.11	≥ 30
New Zealand	Trad	Ewbank	session	8	517	1.77	1.55	2.07	≥ 30
Germany	Sport	French	attempt	8	521	2.08	1.79	2.46	≥ 30
Germany	Sport	French	session	8	483	2.09	1.79	2.49	≥ 30
Germany	Sport	UIAA	attempt	**100**	19,497	2.12	2.06	2.18	≥ 30
Germany	Sport	UIAA	session	**100**	18,435	2.14	2.08	2.21	≥ 30

Table S1: Summary of all 20 Bayesian analyses. n is the number of climbers retained; N is the number of logged ascents on which inference is performed (after per-session aggregation where applicable); $d = e^m$ is the posterior median multiplicative factor in difficulty per grade increment, with the 2.5% and 97.5% quantiles of the posterior shown as endpoints of a 95% central credible interval. The minimum-attempts column lists the per-climber inclusion threshold (set to 30 in all 20 analyses; the minimum-explicit-failures threshold is 1 throughout). n is shown in bold (**100**) where the analysis was limited to the top-100 most-active climbers because more than 100 climbers passed the inclusion thresholds; in the remaining analyses (where n is shown in normal weight), all eligible climbers were used. The slope estimates are mutually consistent across this stratification, suggesting the limit acts as a computational ceiling rather than a substantive analytical choice.

S4 Convergence diagnostics for the main analysis

Table S2 reports convergence diagnostics for the main Australia-Sport per-session analysis (row 4 of Table S1). Diagnostics are reported following Vehtari et al. (2021): rank-normalised split-$\hat{R}$, bulk effective sample size, and counts of divergent transitions and iterations hitting the maximum NUTS treedepth. The slope parameter m converges with $\hat{R} = 1.00$ and $n_{\text{eff}} = 7{,}531$. Across all 6,002 monitored parameters, the maximum $\hat{R}$ is 1.006 and the minimum bulk n_{eff} is 1,103. There were zero divergent transitions and zero iterations hitting the maximum treedepth across all four chains. These diagnostics indicate clean convergence with no need for warmup-length adjustment, step-size retuning, or reparametrisation. The same MCMC settings were used for all 20 analyses.

Diagnostic	**Value**
Number of monitored parameters	6,002
Maximum $\hat{R}$ across all parameters	1.006
Minimum bulk n_{eff} across all parameters	1,103
Median bulk n_{eff} across all parameters	6,077
$\hat{R}$ for slope parameter m	1.000
n_{eff} for slope parameter m	7,531
Posterior mean of m	0.846
Posterior 95% credible interval for m	$[0.829, 0.863]$
Divergent transitions (post-warmup)	0
Iterations hitting max treedepth (post-warmup)	0
Number of chains	4
Iterations per chain	4,000
Warmup iterations per chain	2,000
Total post-warmup draws	8,000
Wall-clock time (4-core parallel)	77.8 minutes

Table S2: Convergence diagnostics for the Australia-Sport per-session analysis. Diagnostics for the every-attempt-logger subset analysis (Section 5.2 of the main text) are similar in quality (max $\hat{R} = 1.01$, min $n_{\text{eff}} = 848$, 0 divergent transitions). Per-analysis diagnostics for the remaining 19 of the 20 analyses summarised in Table S1 are not reported here, because the saved fits from the original submission stored only the posterior draws and not the diagnostics. We have updated the analysis pipeline to capture diagnostics for any future re-runs.

S5 Joint estimation of the Wiener-process step SD

In the 20 main analyses the Wiener-process step SD w is fixed at 0.5 grade units per month, on the grounds that this value encodes a reasonable prior expectation about month-to-month changes in form. To check that the slope estimate m is not sensitive to this choice, we re-fit the Australia-Sport per-session model with w given a $\text{Gamma}(2, 4)$ prior (mean 0.5, mode 0.25, standard deviation ≈ 0.35, with density vanishing at $w = 0$) and jointly estimated alongside m and the climber-grade trajectories. The shape parameter $\alpha = 2$ ensures that the prior density goes to zero linearly as $w \to 0^+$, ruling out the degenerate "no drift" regime that a half-normal or half-Cauchy prior would put high prior mass on. The same MCMC settings as the main analyses were used (4 chains, 4,000 iterations per chain, 2,000 warmup); diagnostics for this run are reported in Table S4.

The joint-estimation posteriors (Table S3) are mutually compatible with the main fixed-w result: the posterior on m shifts only marginally, from 0.846 $[0.829, 0.863]$ (fixed $w = 0.5$) to 0.841 $[0.825, 0.858]$ (joint estimation), with credible intervals that overlap by more than 90% of their width. The posterior on w has median 0.39 with 95% credible interval $[0.36, 0.43]$, indicating that the data can be explained by slightly less monthly drift than the fixed value $w = 0.5$. Because the slope estimate is essentially unaffected by this difference, we retain the fixed-w formulation for the 20 main analyses and present the joint-estimation result here purely as a robustness check.

	median	**mean**	**2.5%**	**97.5%**	n_{eff}
m	0.841	0.841	0.825	0.858	7,300
w	0.392	0.392	0.357	0.426	255
$d = e^m$	2.319	2.319	2.281	2.359	7,300

Table S3: Posterior summaries from the joint-estimation fit to the Australia-Sport per-session data set. w has the lowest effective sample size of any monitored parameter, reflecting slow mixing of the variance.

Diagnostic	**Value**
Number of monitored parameters	6,003
Maximum $\hat{R}$ across all parameters	1.02
Minimum bulk n_{eff} across all parameters	255 (w)
Median bulk n_{eff} across all parameters	8,521
$\hat{R}$ for slope parameter m	1.000
n_{eff} for slope parameter m	7,300
$\hat{R}$ for Wiener-process SD w	1.020
n_{eff} for Wiener-process SD w	255
Divergent transitions (post-warmup)	0
Iterations hitting max treedepth (post-warmup)	2,753
Wall-clock time (4-core parallel)	5.0 hours

Table S4: Convergence diagnostics for the joint-estimation fit. Compared with the fixed-w main run (Table S2), $\hat{R}$ is slightly elevated for w (1.02 vs the ≤ 1.006 ceiling in the fixed-w run) and the sampler hit the maximum NUTS treedepth on 34% of post-warmup iterations, reflecting more challenging posterior geometry when w is a free parameter. The slope m nonetheless converged to high quality and the joint posterior is consistent with the fixed-w result.

S6 Prior sensitivity for the slope parameter

The main analyses use a weakly informative log-normal prior on the slope, $m \sim \text{LogNormal}(-0.5, 0.6^2)$, with central 95% mass on $m \in [0.19, 1.97]$. To check that the main posterior is not driven by this prior choice, we re-fit the Australia-Sport per-session model (with $w = 0.5$ fixed, all other settings as in the main run) under three alternative priors on m: a tighter log-normal with the same median, a substantially wider log-normal with the same median, and a Gamma distribution from a different family. Posterior summaries are reported in Table S5.

The four posteriors agree to three decimal places on the median of m and on the median of

$d = e^m$, and the 95% credible intervals overlap by more than 99% of their width. All four runs converged cleanly with $\hat{R} \leq 1.000$ for m, $n_{\text{eff}}(m) \geq 7{,}531$, and zero divergent transitions. We conclude that, with $N \approx 50{,}000$ ascents informing m, the prior on m has no detectable effect on the posterior, and the main result is determined by the data.

Label	**Prior on m**	**95% prior on m**	m median	2.5%	97.5%	d median	$\hat{R}$
main	LogNormal$(-0.5, 0.6^2)$	$[0.19, 1.97]$	0.846	0.829	0.863	2.330	1.000
tight	LogNormal$(-0.5, 0.3^2)$	$[0.34, 1.09]$	0.845	0.828	0.862	2.328	1.000
wide	LogNormal$(-0.5, 1.0^2)$	$[0.085, 4.31]$	0.846	0.828	0.863	2.329	1.000
gamma	Gamma$(2, 4)$	$[0.06, 1.39]$	0.845	0.829	0.863	2.329	1.000

Table S5: Prior sensitivity for the slope parameter m, fit on the Australia-Sport per-session data set ($n = 100$, $N = 48{,}679$). The first row reproduces the main result. The remaining rows differ only in the prior on m. The posteriors are mutually indistinguishable beyond the third decimal place and the 95% credible intervals overlap nearly completely; the data dominate the prior.

References

Vehtari, A., Gelman, A., Simpson, D., Carpenter, B., and Bürkner, P.-C. (2021). Rank-normalization, folding, and localization: An improved $\hat{R}$ for assessing convergence of MCMC (with discussion). *Bayesian Analysis*, 16(2):667–718.